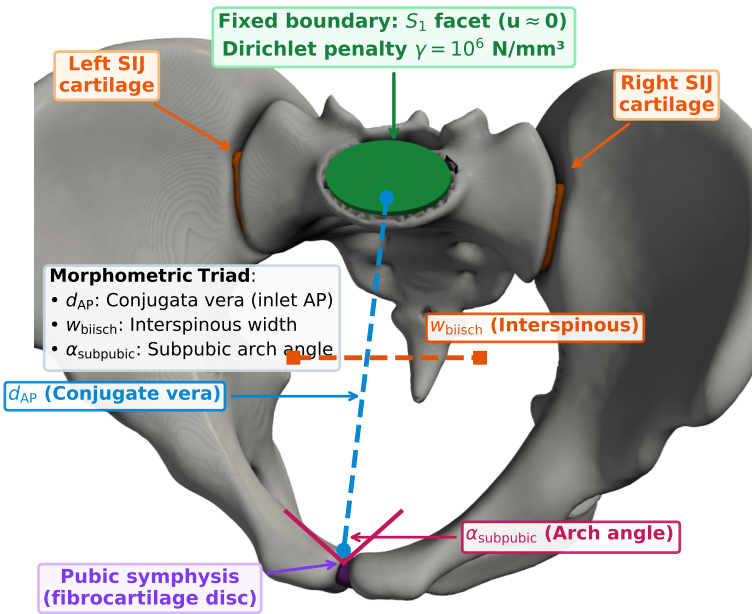
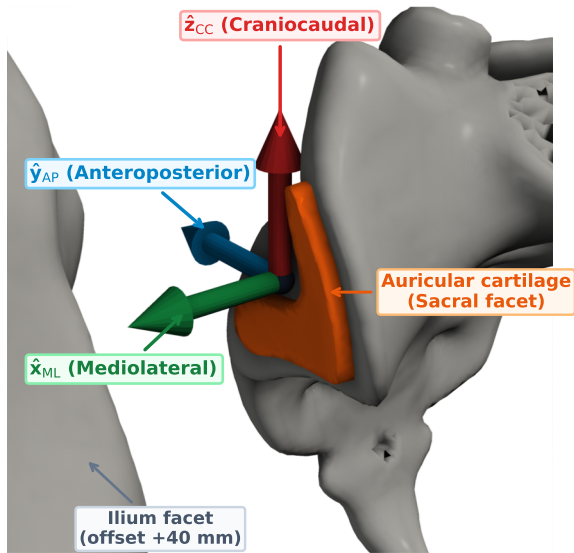


A Pelvic FE assembly & landmarks



B₁ SIJ articulation & triad



B₂ SIJ 6 degrees of freedom (6 DOFs)

3 Rotational DOFs (Fixed-axis xyz):

● θ_{nut} (α_x , ML axis): ML rotation

Absolute ML rotation; nutation sign not inferred

● θ_{abd} (β_y , AP axis): AP rotation

Frontal rotation; side-specific signed convention

● θ_{rot} (γ_z , CC axis): Axial Rotation/Torsion

Rotation about the pelvis-aligned CC axis

3 Translational DOFs (Contact Point p_c):

■ d_{ML} (dx , ML): ML component

ML component; not articular-normal joint gapping

■ d_{AP} (dy , AP): Anteroposterior Gliding/Shear

AP component in the PCA reporting frame

■ d_{CC} (dz , CC): Vertical Craniocaudal Shear

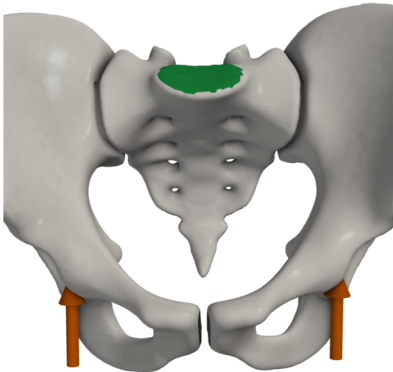
CC component in the PCA reporting frame

$$|\theta| = \frac{1}{2}(\|\theta_L\| + \|\theta_R\|), \quad |\mathbf{d}| = \frac{1}{2}(\|\mathbf{d}_L\| + \|\mathbf{d}_R\|), \quad \Delta_{\text{asym}} = \|\|\mathbf{d}_L\| - \|\mathbf{d}_R\|\|$$

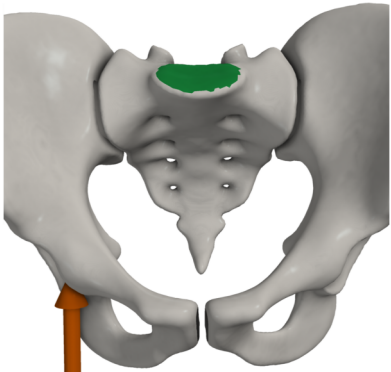
Standardized Standing Loads

Independent Localized Force Pairs

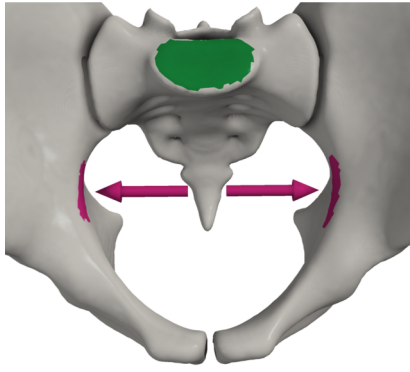
C₁ SP2leg



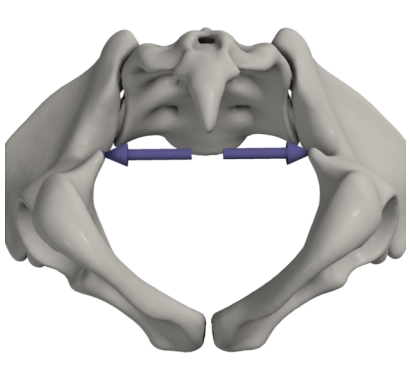
C₂ SP1leg



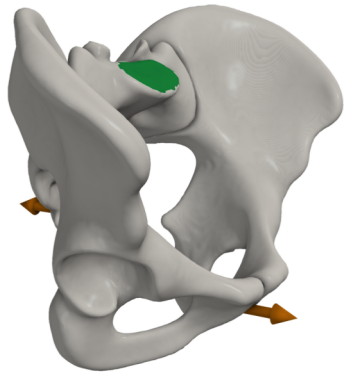
D₁ LAB₁ (Inlet)



D₂ LAB₂ (Midpelvis)



D₃ LAB₃ (Outlet)



Bilateral Standing

2 × +400 N vertical

Acetabular notches

800 N net vertical force

Unilateral Standing

1 × +800 N vertical

Right acetabulum

800 N net vertical force

Internal Ring Pair

±400 N lateral pair

Inner pectineal ring

Zero net force; ML pair

Ischial Force Pair

±400 N lateral pair

Ischial tuberosities

Zero net force; ML pair

Outlet AP Force Pair

±400 N AP pair

Pubis & SCJ

Zero net force; AP pair